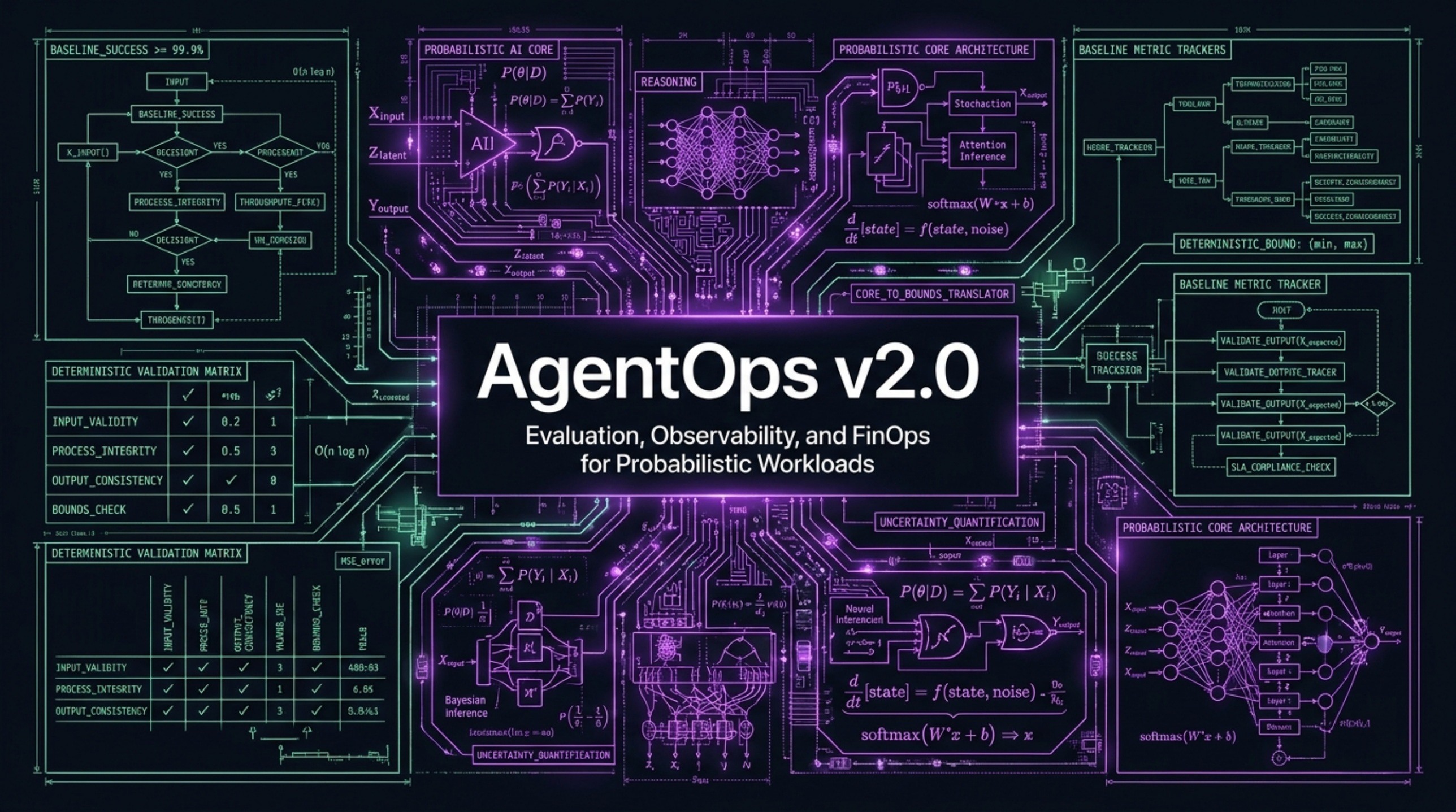
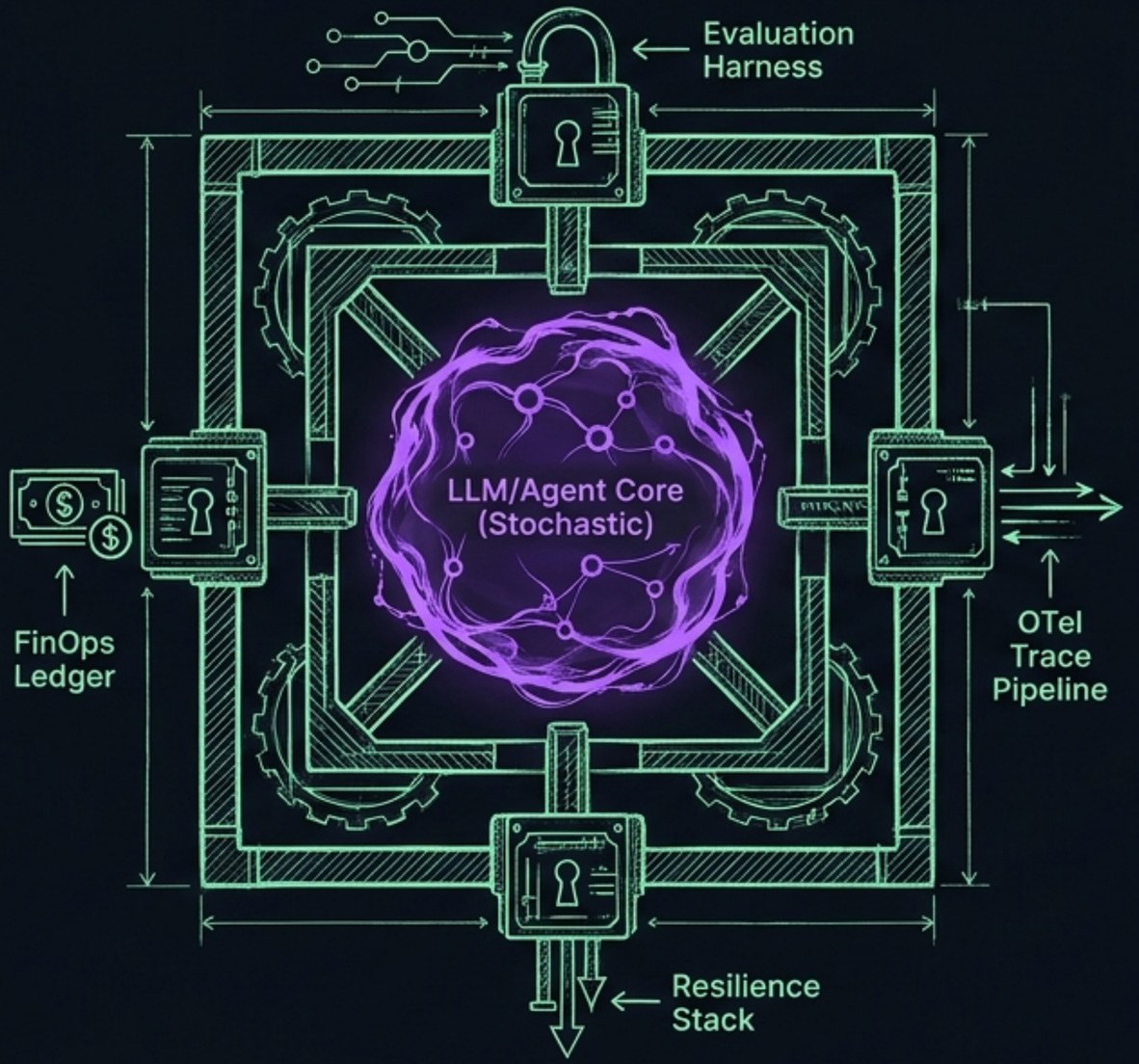
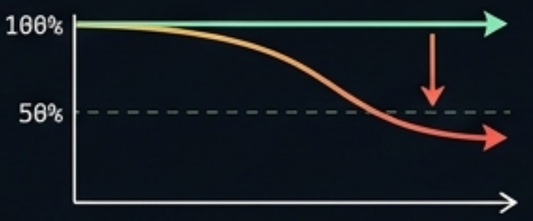

BASELINE_SUCCESS >= 99.9%
INPUT
BASELINE_SUCCESS
PROCESS_INTEGRITY
THROUGHPUT_CHECK
DETERMINISTIC_CONSISTENCY
THROUGHPUT
PROBABILISTIC AI CORE
$P(\theta|D)$
$P(\theta|D) = \sum_{n} P(Y_i)$
X_{input}
Z_{latent}
Y_{output}
AII
REASONING
PROBABILISTIC CORE ARCHITECTURE
Stochaction
Attention Inference
$\text{softmax}(W * \mathbf{x} + b)$
$\frac{d}{dt}[state] = f(state, noise)$
BASELINE METRIC TRACKERS
DETERMINISTIC_BOUND: (min, max)
CORE_TO_BOUNDS_TRANSLATOR
BASELINE METRIC TRACKER
VALIDATE_OUTPUT(X_expected)
VALIDATE_DOTPITE_TRACER
VALIDATE_OUTPUT(X_expected)
VALIDATE_OUTPUT(X_expected)
SLA_COMPLIANCE_CHECK
DETERMINISTIC VALIDATION MATRIX
INPUT_VALIDITY ✓ 0.2 1
PROCESS_INTEGRITY ✓ 0.5 3
OUTPUT_CONSISTENCY ✓ ✓ 0
BOUNDS_CHECK ✓ 0.5 1
O(n log n)
AgentOps v2.0
Evaluation, Observability, and FinOps
for Probabilistic Workloads
DETERMINISTIC VALIDATION MATRIX
MSE_error
INPUT_VALIDITY ✓ ✓ ✓ 3 ✓
PROCESS_INTEGRITY ✓ ✓ ✓ 1 ✓ 6.85
OUTPUT_CONSISTENCY ✓ ✓ ✓ 3 ✓
Bayesian inference
UNCERTAINTY_QUANTIFICATION
Neural inference
$P(\theta|D) = \sum_{out} P(Y_i \mid X_i)$
$\frac{d}{dt}[\text{state}] = f(\text{state}, \text{noise})$
$\text{softmax}(W^{*}x + b) \Rightarrow x$
PROBABILISTIC CORE ARCHITECTURE
Attention
$\text{softmax}(W^{*}x + b)$

Probabilistic Cores Require Deterministic Containment



Classical DevOps tiraes Fails why:

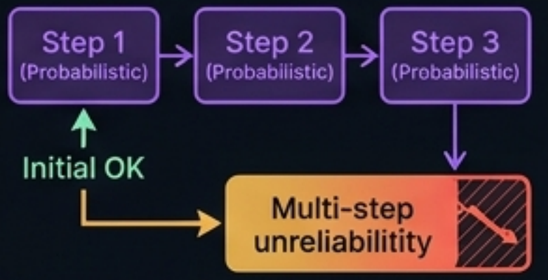
Silent Quality Regressions



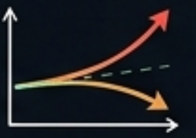
20% drops in factual faithfulness while HTTP endpoints still return 200 OK.



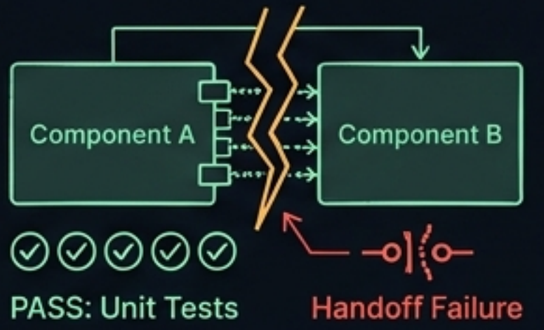
Compounding Failure



Multi-step unreliability across sequential workflows.

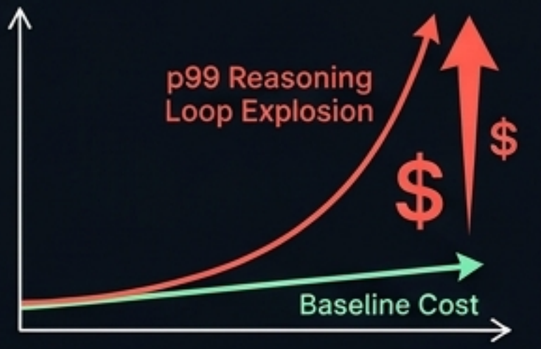


Integration Seams



Passing unit tests but failing at the handoff.

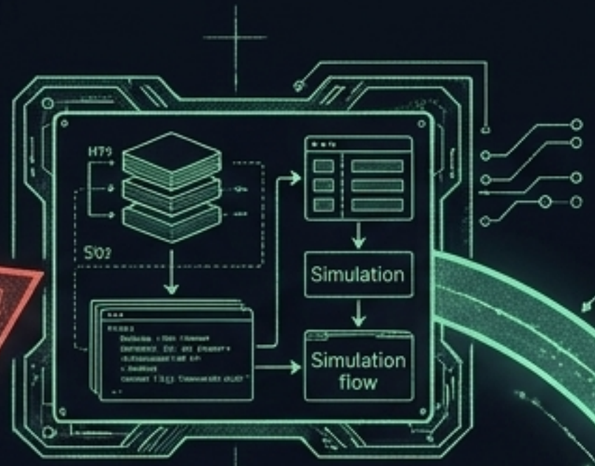
Heavy-Tail Costs



Unbounded economic explosions from p99 reasoning loops.



The AgentOps Lifecycle Flows Through Five Deterministic Pillars

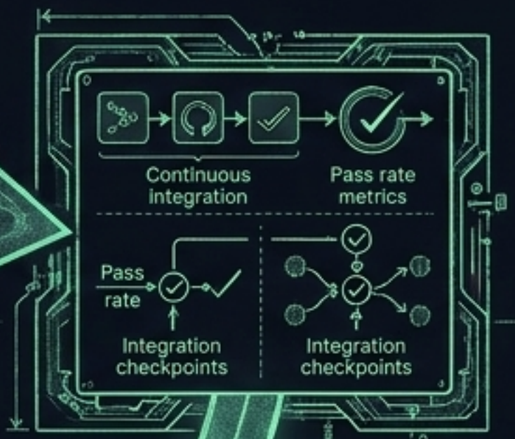


1. Specify & Simulate (Offline)

Curate stratified golden datasets and run 2-Step Agent Simulation.

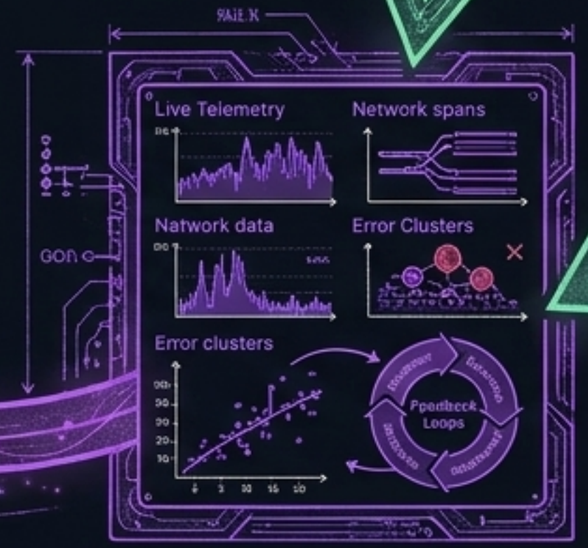
2. Verify Seams (CI/CD)

Gate releases on pass^k consistency and Level 3 integration tests.



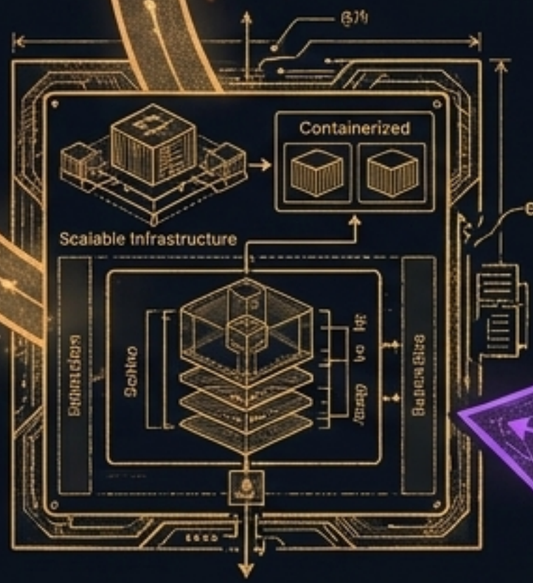
3. Observe & Optimize (Live)

Instrument OpenTelemetry **gen_ai.*** spans, cluster failures, trigger LM Feedback Flywheel.



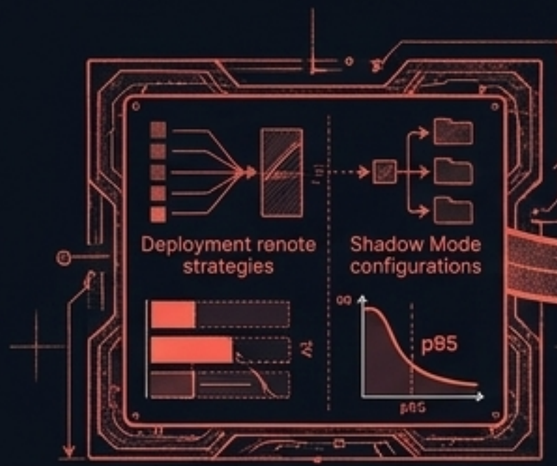
4. Scale & Isolate (Infra)

Match Agent Runtime/GKE workloads and wrap hops in a 4-layer resilience stack.



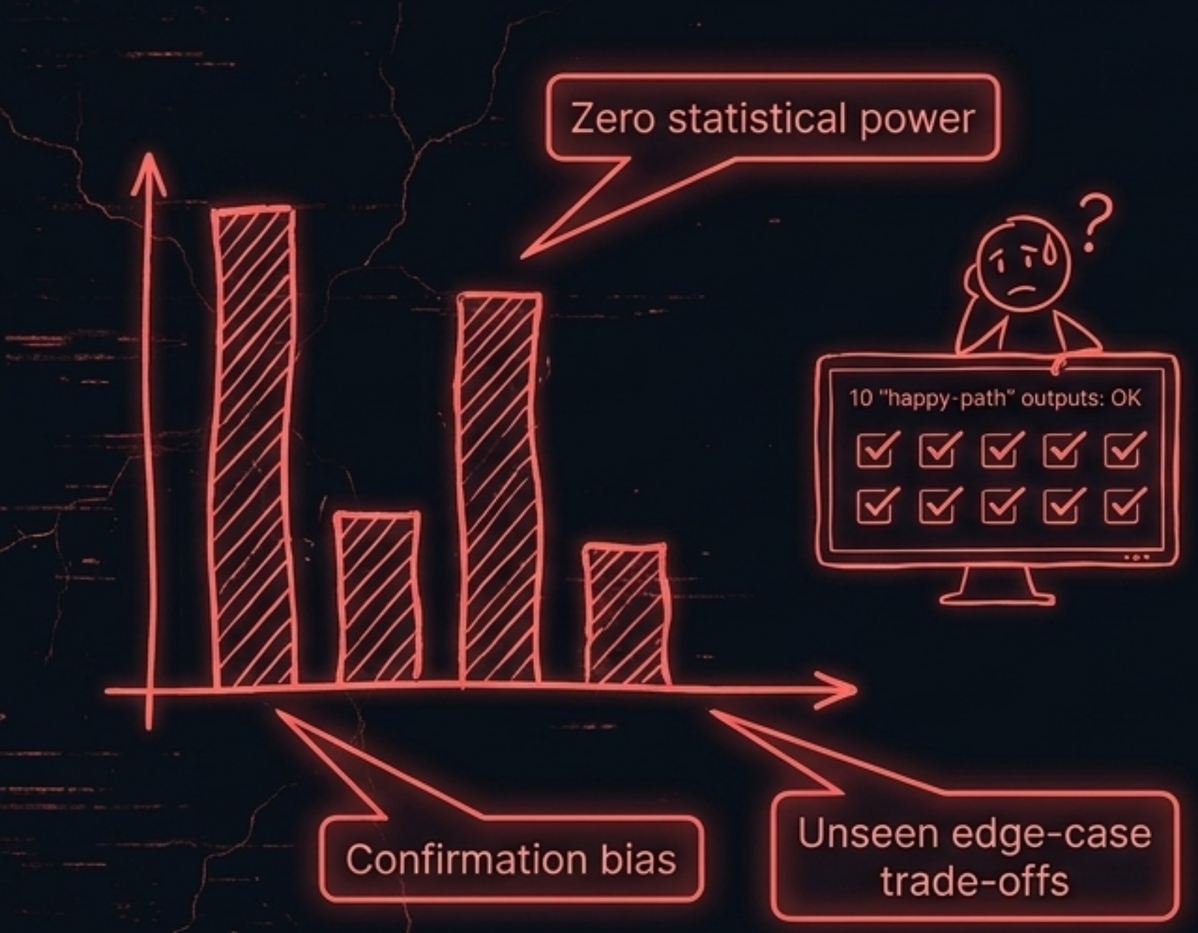
5. Roll Out & Price (FinOps)

Promote via read-only Shadow Mode and price the p95 heavy tail.

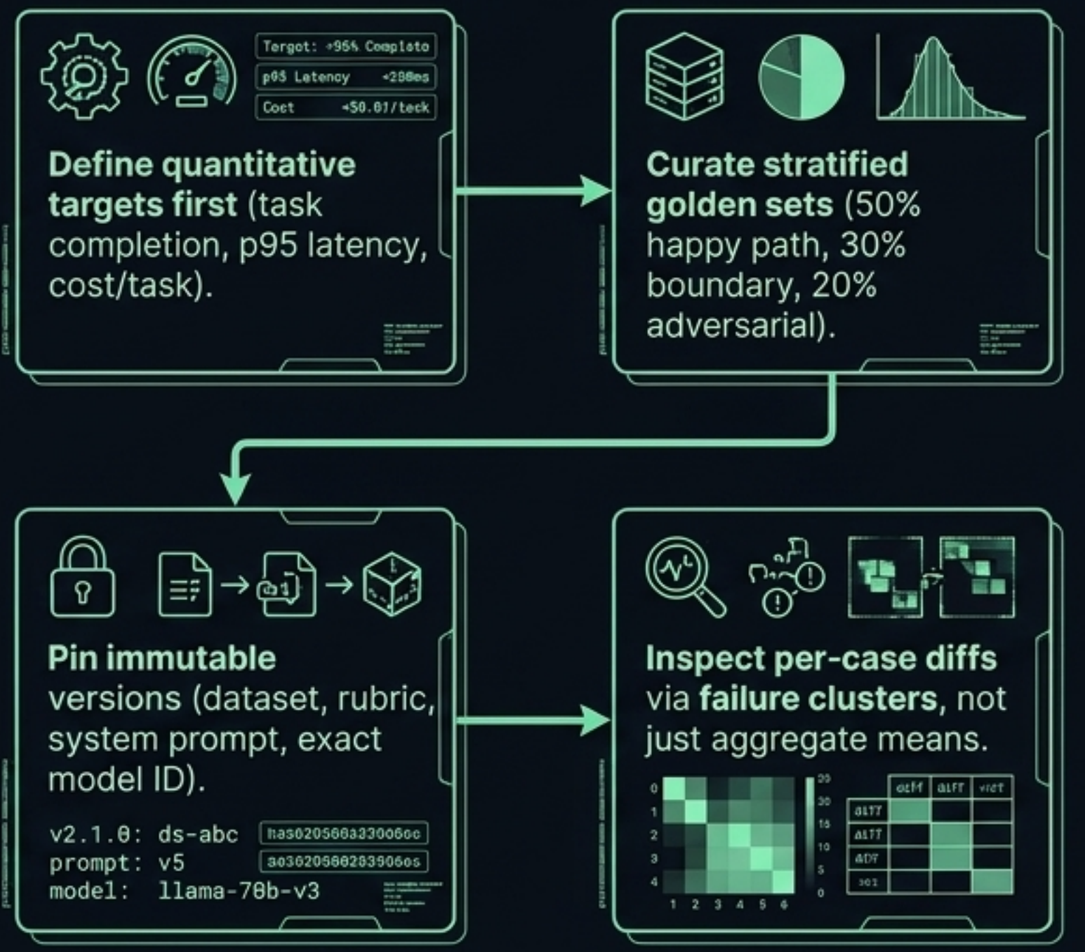


Vibe Checks Mask Regressions at Production Scale

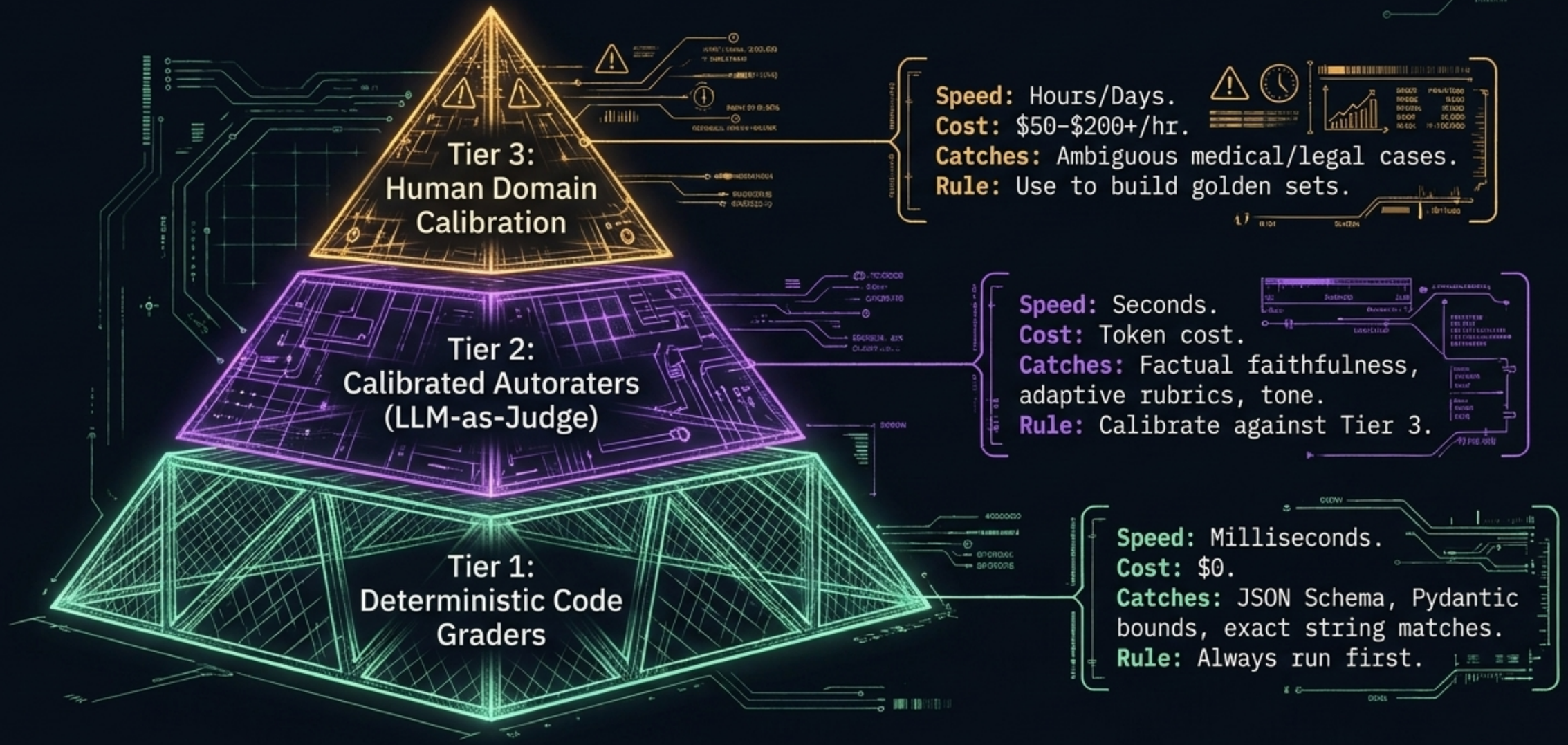
The Vibe Check



Eval-Driven Development (EDD)



Escalate Through the Three Tiers of Evaluation

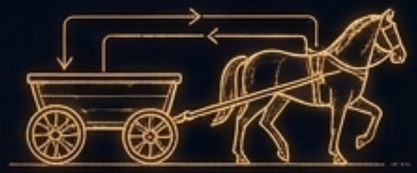


Speed: Hours/Days.
Cost: $50–$200+/hr.
Catches: Ambiguous medical/legal cases.
Rule: Use to build golden sets.

Speed: Seconds.
Cost: Token cost.
Catches: Factual faithfulness, adaptive rubrics, tone.
Rule: Calibrate against Tier 3.

Speed: Milliseconds.
Cost: $0.
Catches: JSON Schema, Pydantic bounds, exact string matches.
Rule: Always run first.

Uncalibrated Autoraters Introduce Systematic Measurement Errors

Bias Type	Visual Metaphor	Production Mitigation
Verbosity Bias		Grade atomic factual precision/recall; penalize verbosity.
Positional Bias		Run pairwise comparisons twice, swapping orders (A,B then B,A).
Self-Preference Bias		Grade with a different/higher-tier model family in a fresh context window.
Post-Hoc Rationalization		Enforce Chain-of-Thought BEFORE judgment in structured output.

Human-Autorater Alignment Standard: Cohen's Kappa ($\kappa \geq 0.75$)

Required before automating any CI/CD release gates.

Match the Evaluation Metric to the Architectural Boundary

Computation Metrics

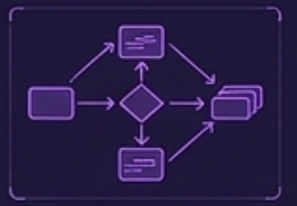
$0 deterministic algorithms like JSON validation or token budgeting.



Gen AI Evaluation Service

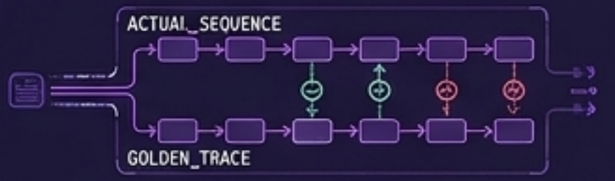
Adaptive Rubrics (Reference-free)

Dynamically generates pass/fail criteria per input for open-ended queries.



Trajectory Metrics (Reference-based)

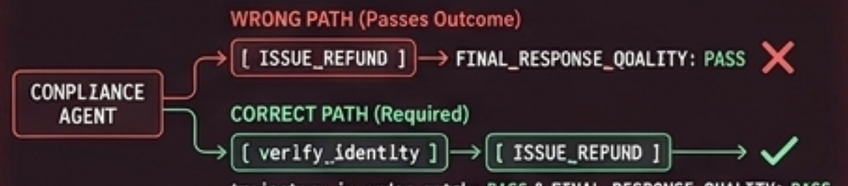
Compares actual tool sequence to a golden trace. (`trajectory_exact_match`, `trajectory_in_order_match`).



The Right Answer, Wrong Path Trap

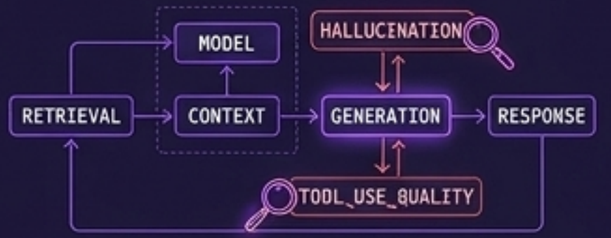
Grading a compliance agent solely on outcome (`FINAL_RESPONSE_QUALITY`) passes an agent that issues refunds without calling `verify_identity`.

Always pair outcome rubrics with `trajectory_in_order_match`.

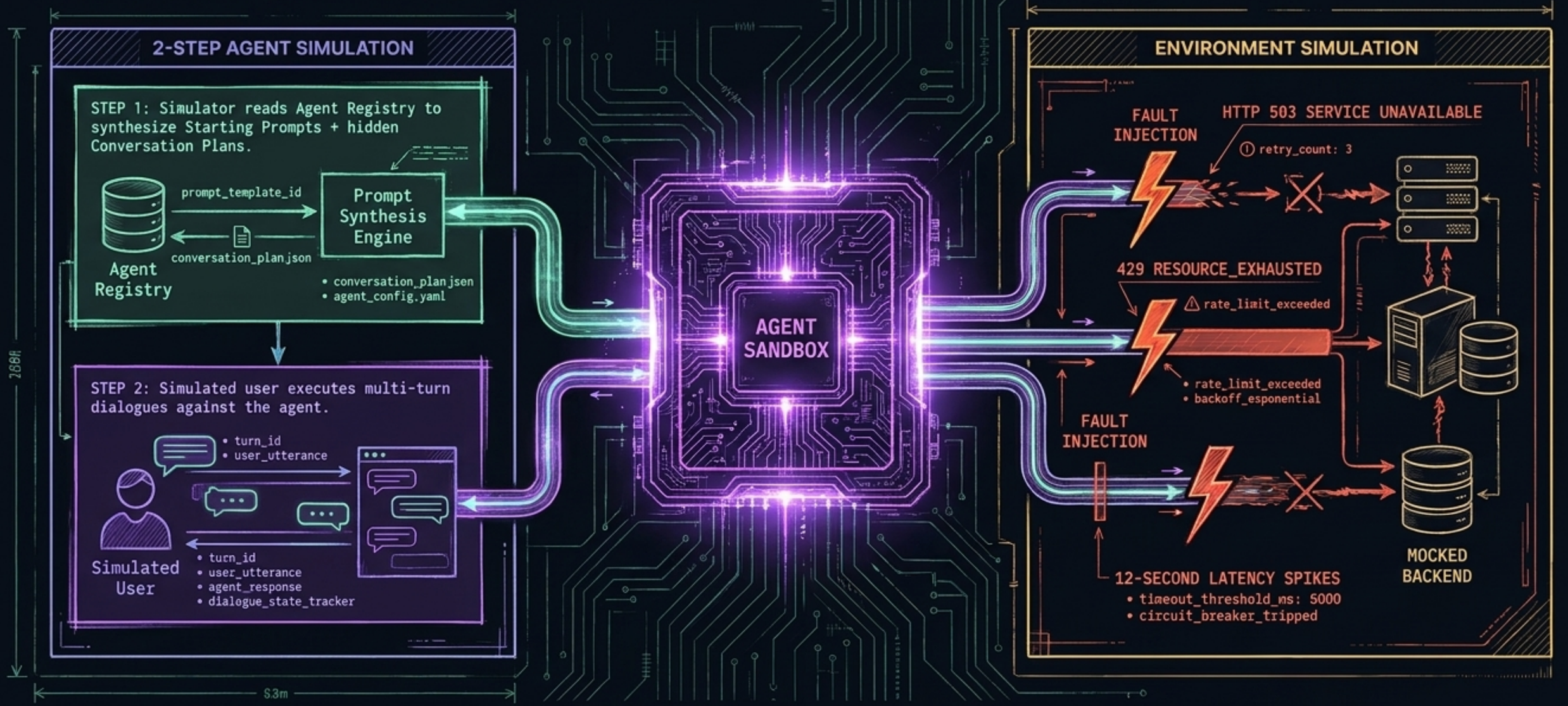


Grounding Autoraters

Inspects RAG traces for `HALLUCINATION` and `TOOL_USE_QUALITY`.



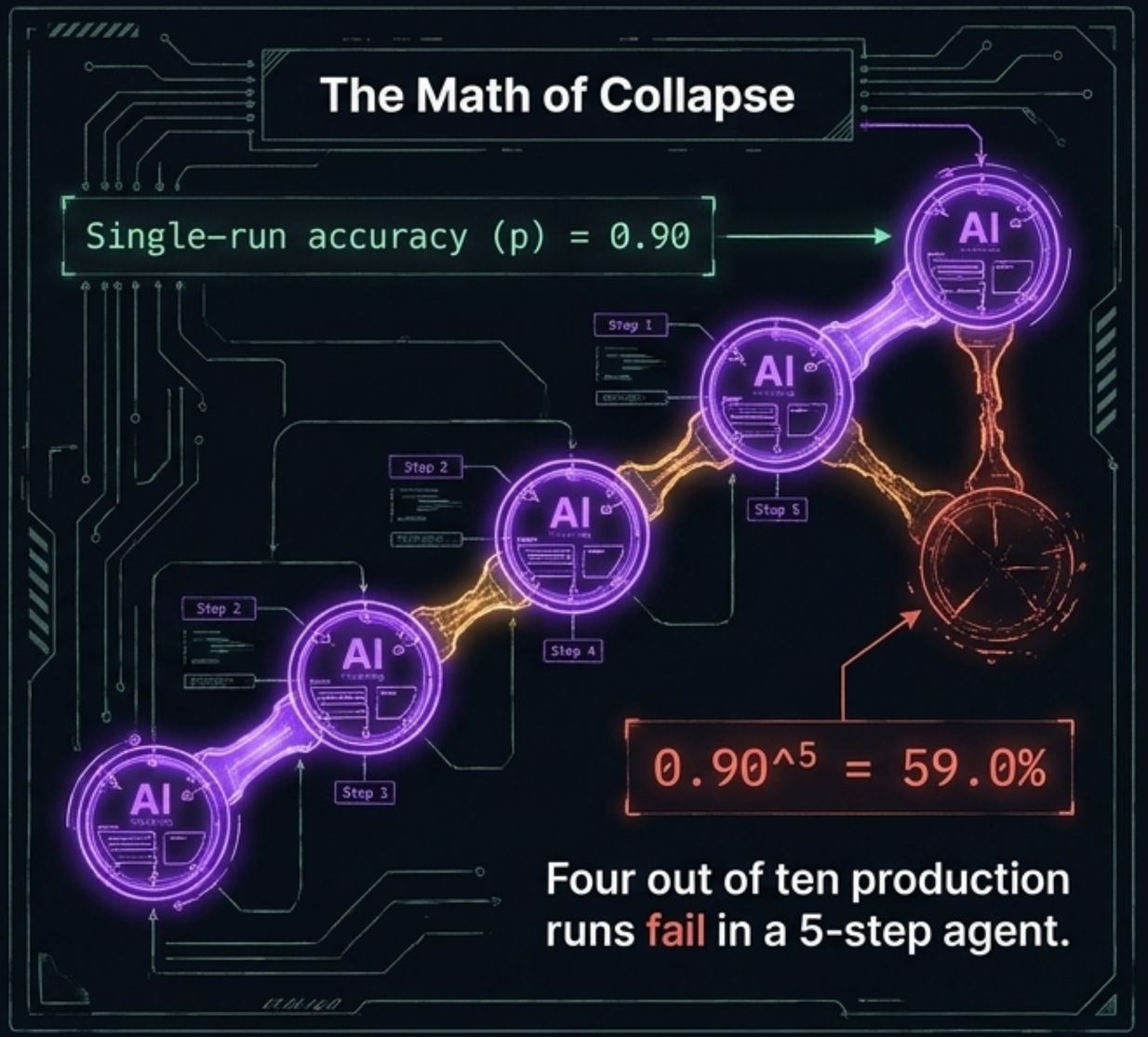
Simulate Multi-Turn Environments to Catch Edge Cases Before Launch



Proves retry/fallback logic works without touching live production APIs.

Multi-Step Unreliability Compounds Exponentially

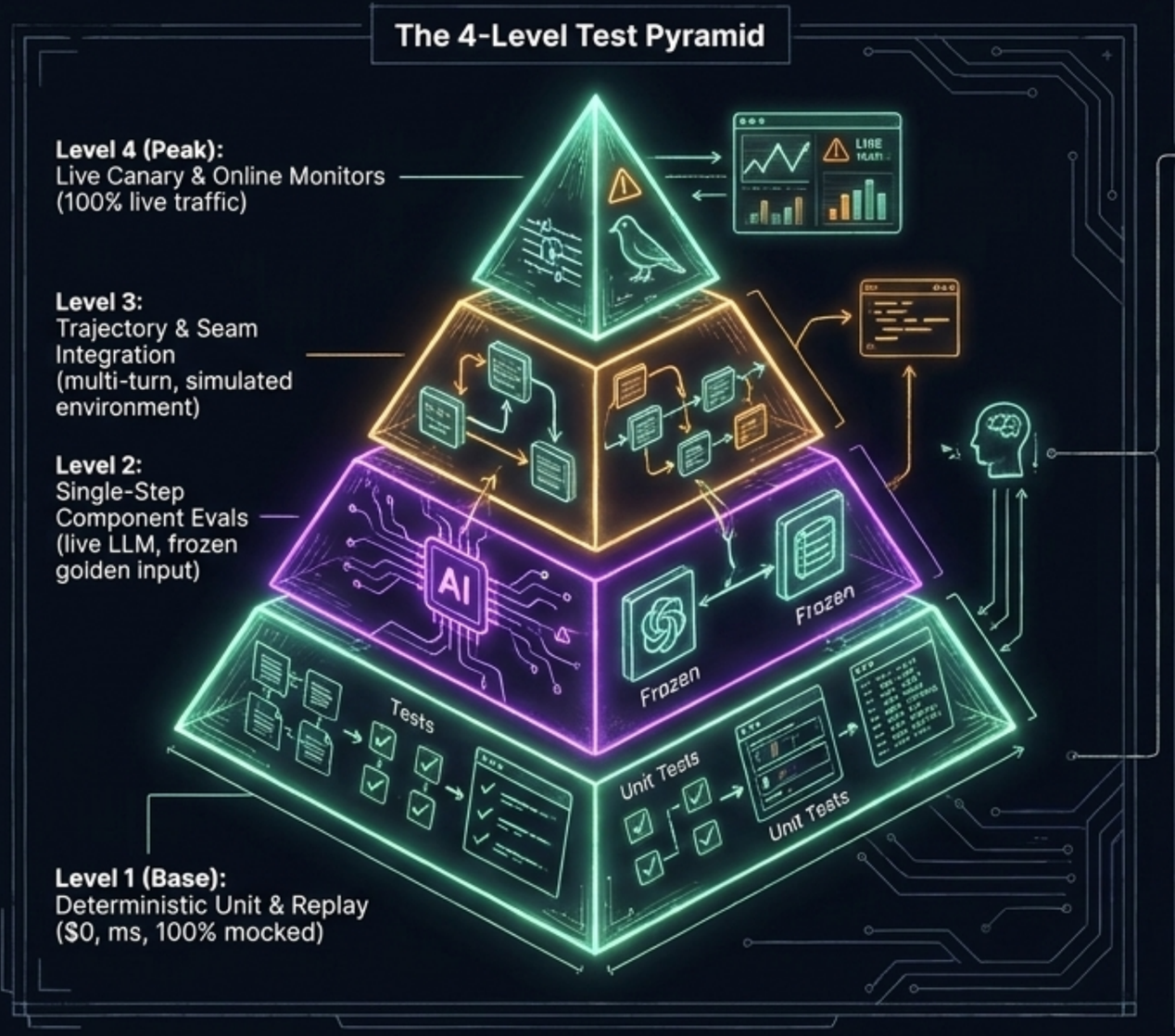




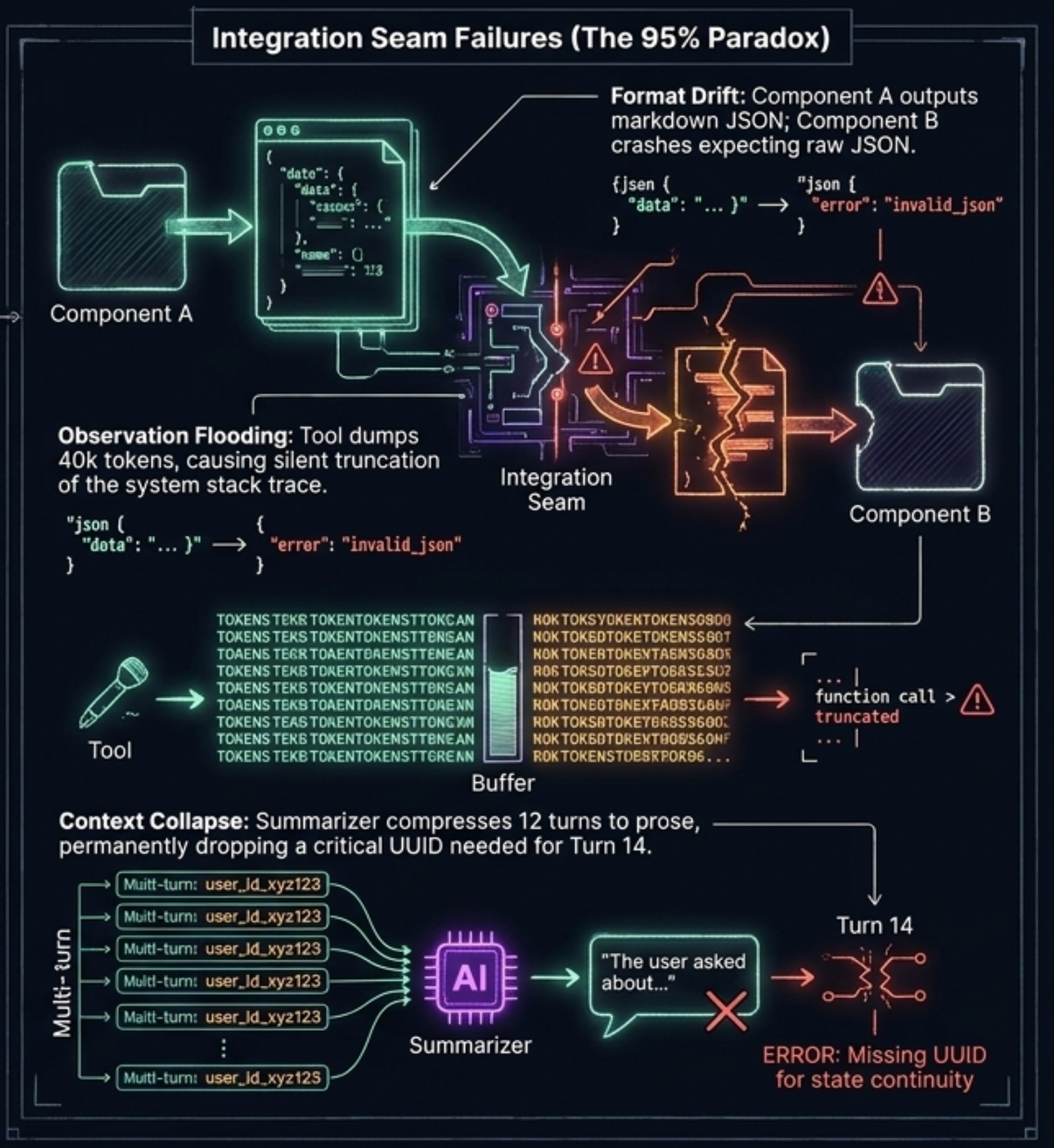
Architectural Rule: Use pass^k for customer-facing agents and mutating workflows. Use pass@k ONLY when a deterministic compiler filters failures automatically.

Ninety-Five Percent Component Success Masks Integration Seam Failures

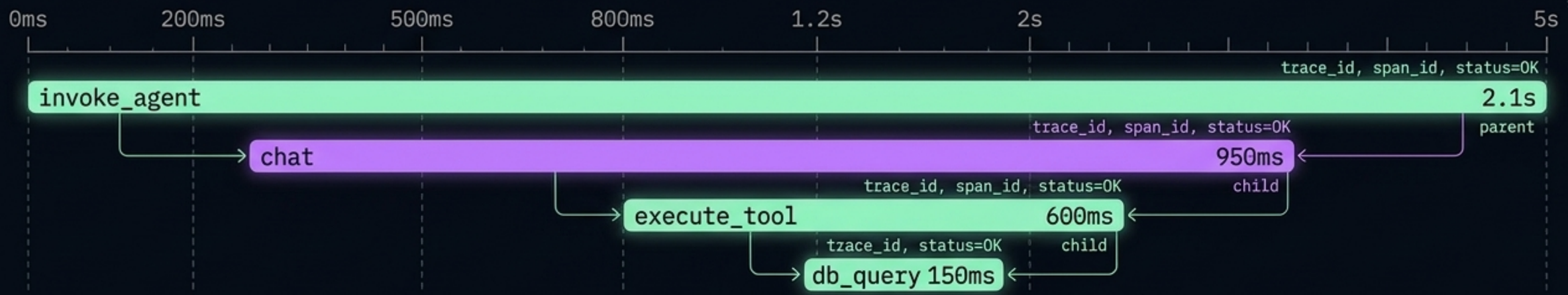
The 4-Level Test Pyramid



Integration Seam Failures (The 95% Paradox)



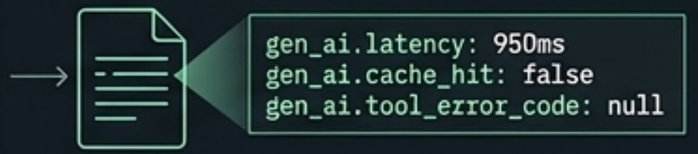
OpenTelemetry Semantic Conventions Expose the Black Box



Telemetry Governance

Metadata-First Default

gen_ai.* attributes capture latencies, cache counts, and tool error codes without logging raw prompts.



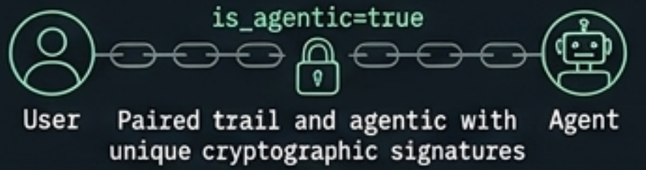
Opt-In Payload Capture

Requires Sensitive Data Protection (SDP) de-identification to scrub PII/secrets before reaching Cloud Logging.



Dual-Attribution Audit

is_agentic=true flag cryptographically pairs the SPIFFE Agent Identity with the human user to prevent forensic blind spots.



Telemetry Signatures Reveal the Root Cause of Production Drift

DIAGNOSTIC MATRIX

	REASONING LOOPS	Symptom: 8-15 oscillating chat/tool spans.	Cause: Ambiguous tool errors. Fix: Structured error hints, enforce max_iterations.
	LATENCY SPIKES	Symptom: p95 jump to 14s, ttft_ms < 1.2s.	Cause: Sequential independent reads. Fix: MCP connection pooling, ParallelAgent fan-out.
	IGNORED TOOLS	Symptom: High 'no-tool interaction' rate.	Cause: Vague overlapping schemas. Fix: Sharpen docstrings, enforce trajectory_in_order_match.

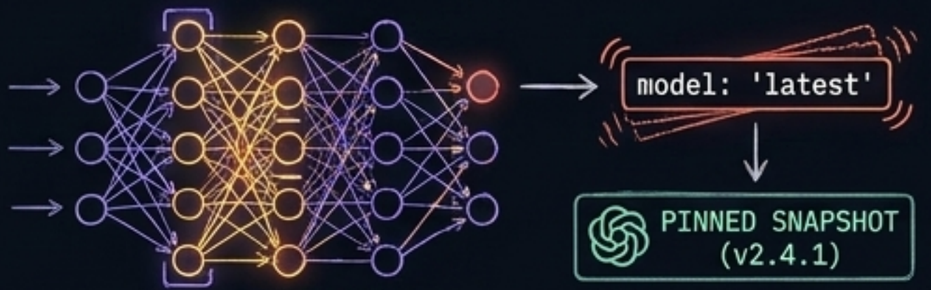
TWO FORMS OF DRIFT

INPUT/DATA DRIFT



World Changes
Real user phrasing shifts; retrieval similarity drops.
Fix: Expand RAG corpus.

MODEL/PROVIDER DRIFT



System Changes
JSON parse errors rise due to unpinned 'latest' aliases.
Fix: Pin immutable model snapshots.

Automate Discovery and Remediation with the Feedback Flywheel



1. Capture

Online Evaluation Monitors sample 1-5% of live traffic and score via reference-free autoraters.

2. Scrub (SDP)

Strict PII removal before storage.

3. Cluster

Failure Cluster Analysis groups raw errors into actionable buckets (e.g., "34% fail on relative dates") ranked by Frequency × Severity.

4. Reproduce

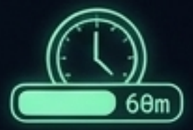
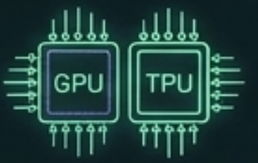
Convert traces to Level 3 trajectory evals - asserting outcomes, not exact transcripts.

5. Optimize

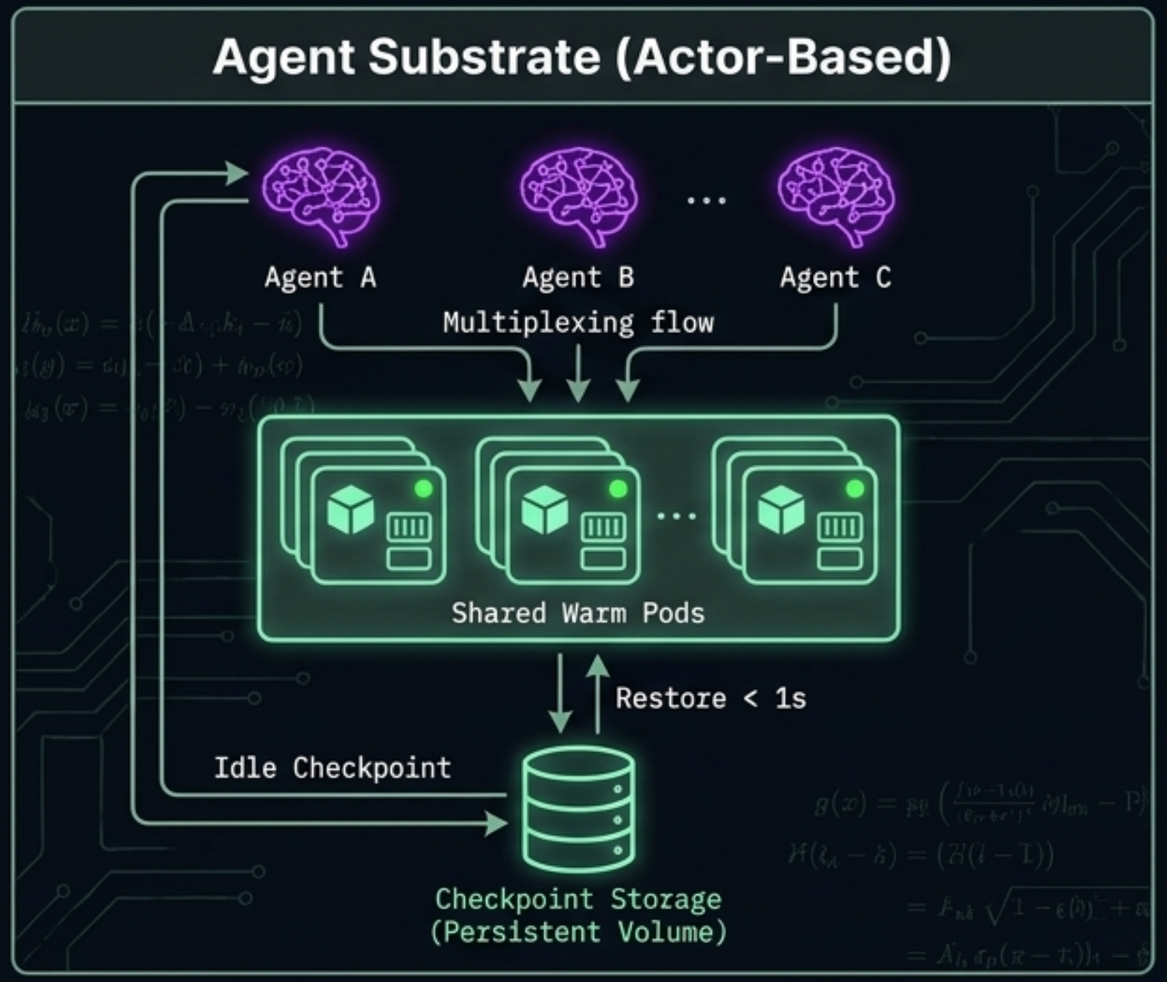
Pass the cluster to the Data-Driven Agent Optimizer.

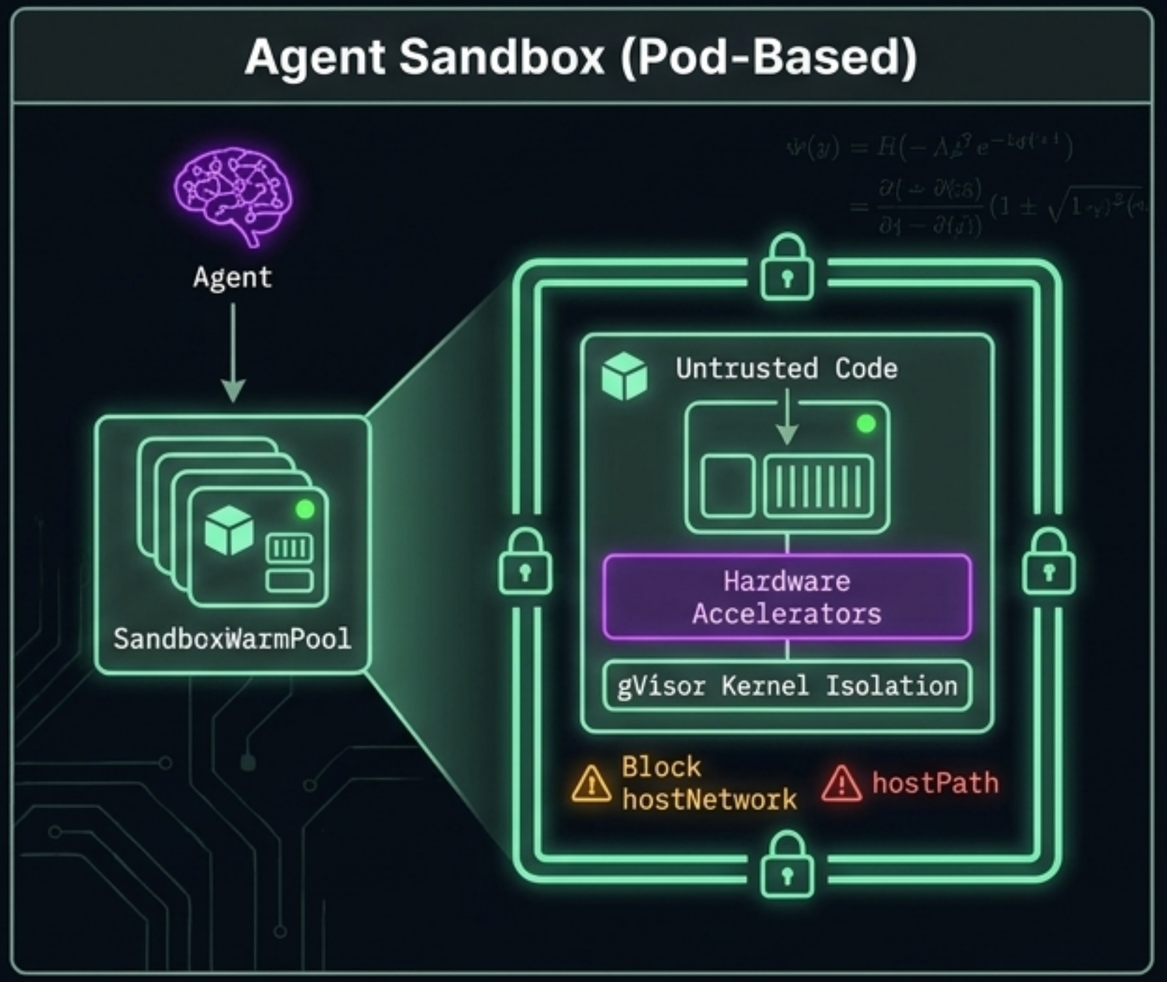
Critical Guardrail: Always maintain a held-out validation split invisible to the optimizer to prevent prompt memorization/overfitting.

Match the Workload to the Google Cloud Runtime

Feature/Trait	Agent Runtime	Cloud Run	GKE
Primary Fit	Best for Stateful Python/ADK orchestrators. 	Best for polyglot agents and Custom Streamable HTTP MCP servers. 	Best for multi-agent meshes and self-hosting open-weight models. 
State & Memory	☑ Built-in Sessions, Memory Bank, and Code Execution sandboxes. 	☒ Ephemeral memory (must externalize state to Memorystore/Firestore). 	☑ Highly stateful volume persistence. 
MCP Hosting	☒ Cannot host standalone custom MCP servers. 	☑ Primary choice for serverless custom HTTP MCPs. 	☑ High-density, private in-cluster MCP fleets. 
GPUs / Limits	☒ Calls managed APIs only. 	HTTP timeouts up to 60 mins. 	☑ Custom GPUs/TPUs for Gemma 4. 

Isolate Multitenant Agents on Kubernetes





- Best for high-density stateful agents that frequently go idle between user turns.
- Idle agents automatically checkpoint memory/files to storage and restore in < 1 second.

- Best for untrusted code execution or hardware accelerators.
- Mandatory gVisor kernel isolation enforced via Validating Admission Policy (blocks hostNetwork/hostPath).

Wrap Probabilistic Endpoints in a Four-Layer Resilience Stack



Layer 1 (Outer Shield): **Backoff & Idempotency:** Exponential backoff with Full Jitter to prevent thundering herd retry storms. Enforce deterministic idempotency_key hashes on mutating tools.

Layer 3 (Third Shield): **Validated Fallbacks:** Cross-region or cross-model routing (e.g., Argon to Flash). Requires fallback-specific prompt configurations to prevent parameter crashes.

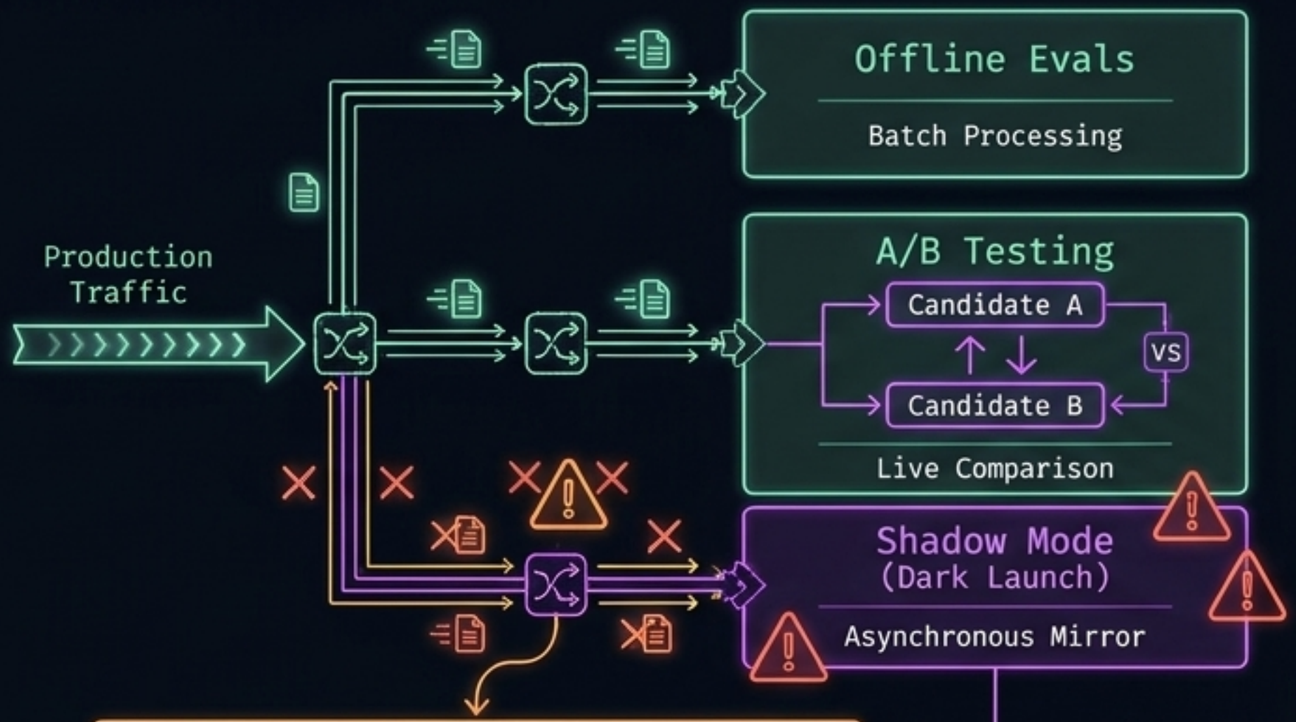
Layer 4 (Inner Core): **Context-Rich Escalation:** Hand off a structured bundle (trace_id, history, verified entities) to a human specialistics before policy boundaries trigger.

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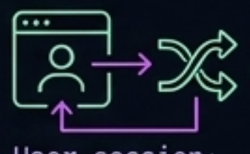
Shadow Mode Requires a Read-Only Tool Sandbox

Traffic Splitting Flowchart

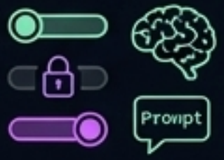


Critical Warning: Live requests duplicate asynchronously to candidate agents. If candidate calls mutating tools (e.g., e.g., refund_order), running unsandboxed shadow mode corrupts production data! **Must enforce a read-only sandbox.**

The 4 Elements of Controlled A/B Tests

1 

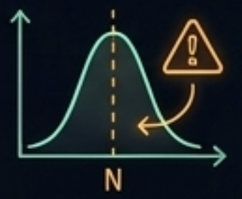
Sticky Randomization Unit: Randomize per user/session, never per HTTP turn (ruins prefix caching & causal attribution).

2 

Single-Variable Isolation: Change prompt OR model, not both.

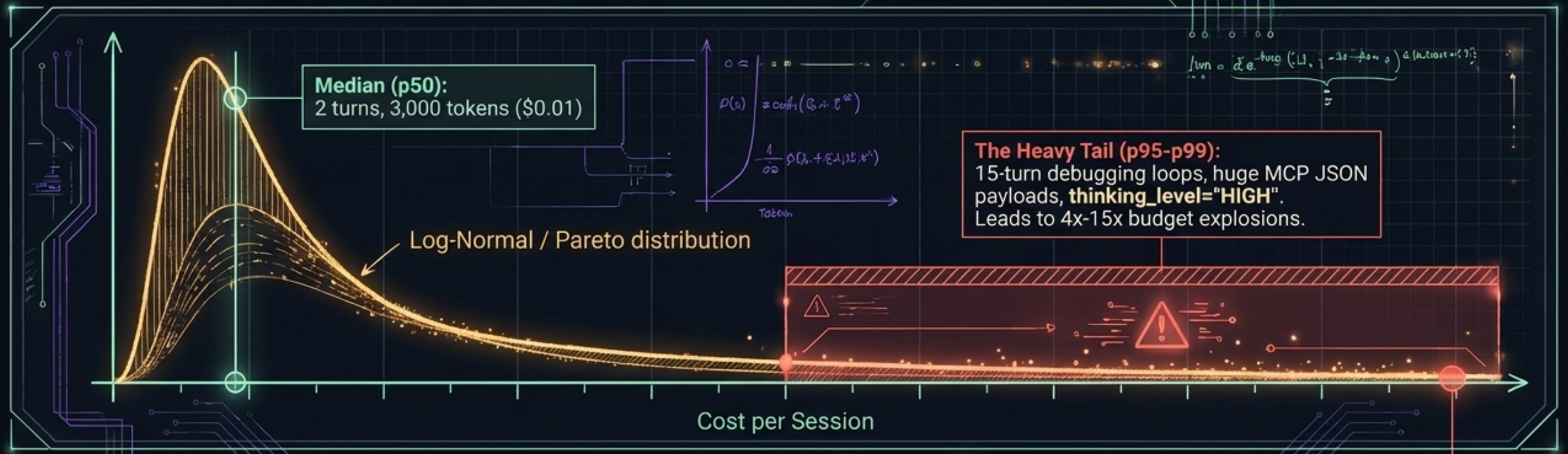
3 

Non-Inferiority Guardrails: Track token cost/task and latency alongside primary success metrics.

4 

Statistical Stopping Rules: No early peeking/p-hacking before predefined sample size N is reached.

Sizing on Median Usage Misses the Expensive Heavy Tail

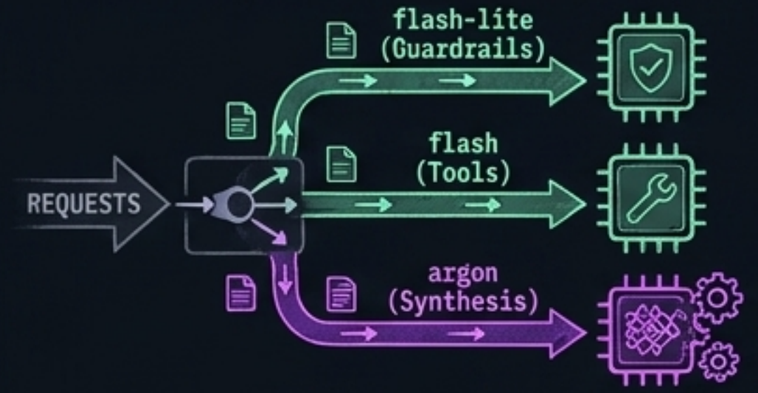


$$O(N^2) \text{ multi-turn session cost} \approx N \cdot S_{\text{static}} + \sum_{t}^{N} (N - t + 1) \cdot \Delta_t$$

FinOps Rule: Always model TPM quotas and budgets using p95/p99-weighted buckets, and clip runaway tails with strict max_turns limits.

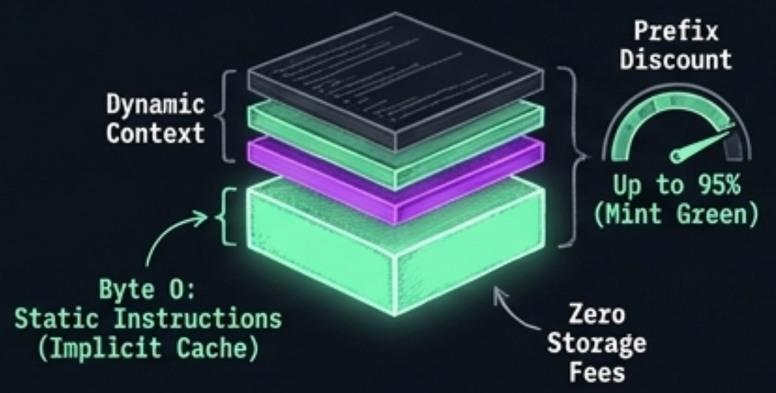
Bend the Token Cost Curve via Architectural Interventions

Tiered Semantic Routing



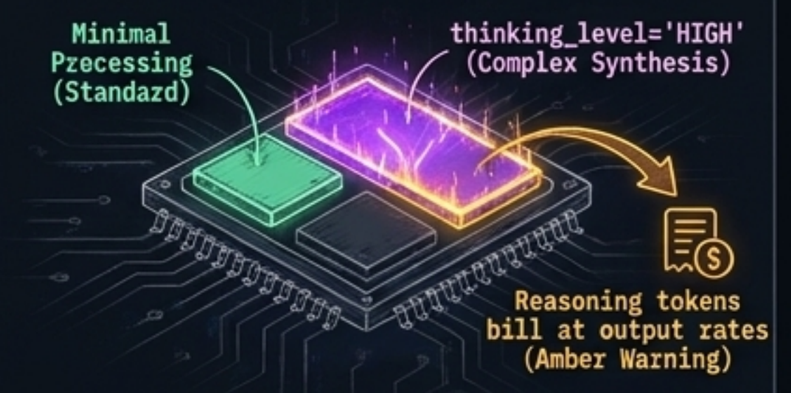
Send guardrails to flash-lite, tools to flash, synthesis to argon. (50-80% savings).

Prefix-Aligned Caching



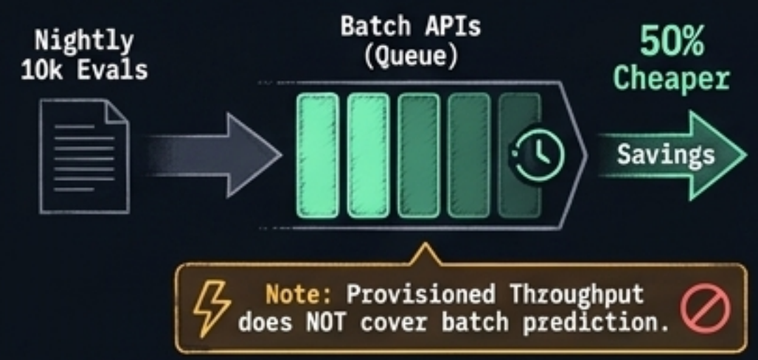
Keep static instructions at byte 0. Implicit caching has zero storage fees. (Up to 95% prefix discount).

Effort Right-Sizing



Reserve thinking_level='HIGH' strictly for complex synthesis; reasoning tokens bill at output rates.

Async Batch / Flex Inference



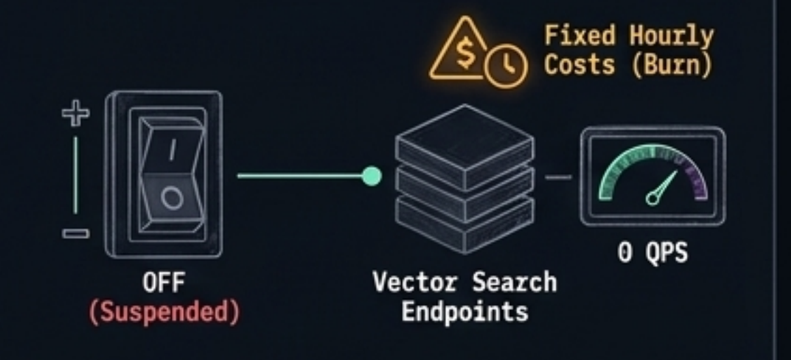
Route nightly 10k evals to Batch APIs (50% cheaper). Note: Provisioned Throughput does NOT cover batch prediction.

Observation Masking



Filter tabular JSON inside sandboxes before returning to context to halt O(N^2) bloat.

Idle Infra Hygiene



Suspend unused Vector Search endpoints (which burn fixed hourly costs even at 0 QPS).

The **Human-Review Tax** Defines Net Enterprise **ROI**

[Gross Savings] – [AI OpEx] – [Human-Review Tax] = NET ROI



Worked Example Breakdown

Baseline: 50k claims @ $12 manual cost = $600k baseline.

Automation: 80% automated = $480k **Gross Savings**. **AI OpEx** = $30k.

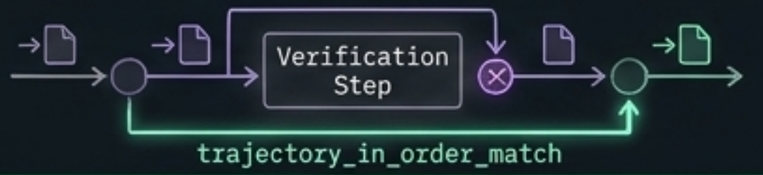
The Trap (Step 4): If compliance requires a 6-minute human audit on all automated claims ($240k) plus specialist rework on subtle AI errors ($75k), the **Human-Review Tax** is $315k.

Result: Net ROI drops from $450k to $135k. ↓

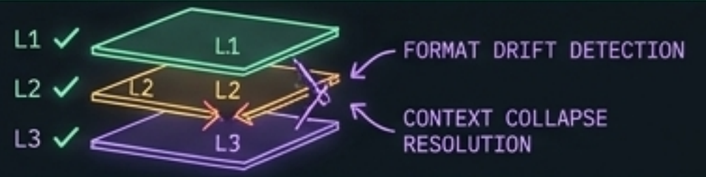
Remediation: Implement $0 Tier 1 deterministic gates and calibrate routing so humans only audit genuinely uncertain slices.

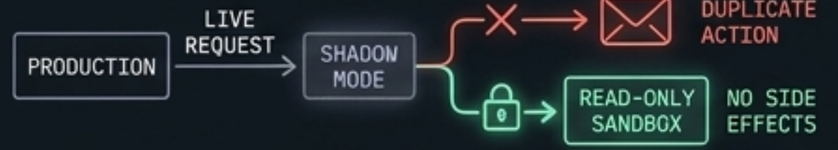
Production Diagnostics Cheat Sheet

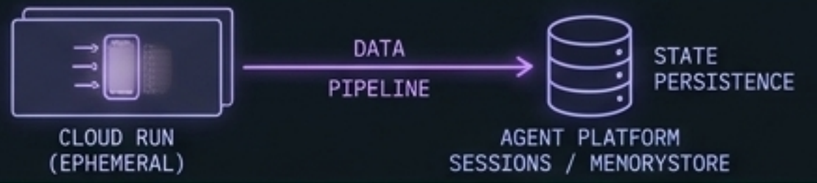
Observed Symptom	Correct AgentOps Remediation
97% Outcome score, but agent skips verification tools.	Enforce trajectory_in_order_match.
"Region temporarily at capacity" error on Agent Runtime.	Physical pool exhausted; move regions (Do not file quota ticket). ✕
Components pass L2 >95% but end-to-end fails 45%.	Level 3 Integration Seam testing (format drift/context collapse).
Shadow mode sends duplicate cancellation emails.	Implement read-only sandbox for live traffic duplication.
Cloud Run agent loses context randomly.	Cloud Run is ephemeral; externalize state to Agent Platform Sessions/Memorystore.











You cannot eliminate variance inside the model. You must bound, measure, optimize, and price it at the boundary.